13.1 Integrating x^n

Integration is the reverse process of differentiation:

Function → multiply by the power → subtract one from the power → **Gradient Function**

$x^n \rightarrow$ multiply by the power $\rightarrow$ subtract one from the power $\rightarrow nx^{n-1}$ — Differentiating x^n

$\frac{x^{n+1}}{n+1} \leftarrow$ divide by the new power $\leftarrow$ add one to the power $\leftarrow x^n$ — Integrating x^n

Constant terms disappear when you differentiate. This means that when you differentiate functions that only differ in the constant term, they will all differentiate to give the same function. To allow for this, you need to add a **constant of integration** at the end of a function when you integrate.

$y = x^2 + 5$

$y = x^2$ → Differentiate → $\frac{dy}{dx} = 2x$ → Integrate → $y = x^2 + c$

$y = x^2 - 19$

c — This is the constant of integration.

- **If $\frac{dy}{dx} = x^n$, then $y = \frac{1}{n+1}x^{n+1} + c$, $n \neq -1$.**
- **If $f'(x) = x^n$, then $f(x) = \frac{1}{n+1}x^{n+1} + c$, $n \neq -1$.**

Links You cannot use this rule if $n = -1$ because $\frac{1}{n+1} = \frac{1}{0}$ and so is not defined.
You will learn how to integrate the function x^{-1} in Year 2. → Year 2, Section 11.2

Example 1

Find y for the following:

a $\frac{dy}{dx} = x^4$ **b** $\frac{dy}{dx} = x^{-5}$

a $y = \frac{x^5}{5} + c$

b $y = \frac{x^{-4}}{-4} + c = -\frac{1}{4}x^{-4} + c$

Use $y = \frac{1}{n+1}x^{n+1} + c$ with $n = 4$.
Don't forget to add c.

Remember, adding 1 to the power gives $-5 + 1 = -4$.
Divide by the new power (-4) and add c.

Example 2

Find $f(x)$ for the following:

a $f'(x) = 3x^{\frac{1}{2}}$ **b** $f'(x) = 3$

a $f(x) = 3 \times \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}} + c = 2x^{\frac{3}{2}} + c$

Remember $3 \div \frac{3}{2} = 3 \times \frac{2}{3} = 2$

Simplify your answer.

b $f'(x) = 3 = 3x^0$

So $f(x) = 3 \times \frac{x^1}{1} + c = 3x + c$

$x^0 = 1$, so 3 can be written as $3x^0$.

You can integrate a function in the form kx^n by integrating x^n and multiplying the integral by k.

- **If $\frac{dy}{dx} = kx^n$, then $y = \frac{k}{n+1}x^{n+1} + c$, $n \neq -1$.**
- **Using function notation, if $f'(x) = kx^n$, then $f(x) = \frac{k}{n+1}x^{n+1} + c$, $n \neq -1$.**
- **When integrating polynomials, apply the rule of integration separately to each term.**

Watch out You don't need to multiply the constant term (c) by k.

Example 3

Given $\frac{dy}{dx} = 6x + 2x^{-3} - 3x^{\frac{1}{2}}$, find y.

$y = \frac{6x^2}{2} + \frac{2}{-2}x^{-2} - \frac{3}{\frac{3}{2}}x^{\frac{3}{2}} + c$

$= 3x^2 - x^{-2} - 2x^{\frac{3}{2}} + c$

Apply the rule of integration to each term of the expression and add c.

Now simplify each term and remember to add c.

Exercise 13A

1 Find an expression for y when $\frac{dy}{dx}$ is the following:

a x^5 **b** $10x^4$ **c** $-x^{-2}$ **d** $-4x^{-3}$ **e** $x^{\frac{2}{3}}$ **f** $4x^{\frac{1}{2}}$

g $-2x^6$ **h** $x^{-\frac{1}{2}}$ **i** $5x^{-\frac{3}{2}}$ **j** $6x^{\frac{1}{3}}$ **k** $36x^{11}$ **l** $-14x^{-8}$

m $-3x^{-\frac{2}{3}}$ **n** -5 **o** $6x$ **p** $2x^{-0.4}$

2 Find y when $\frac{dy}{dx}$ is given by the following expressions. In each case simplify your answer.

a $x^3 - \frac{3}{2}x^{-\frac{1}{2}} - 6x^{-2}$ **b** $4x^3 + x^{-\frac{2}{3}} - x^{-2}$ **c** $4 - 12x^{-4} + 2x^{-\frac{1}{2}}$

d $5x^{\frac{2}{3}} - 10x^4 + x^{-3}$ **e** $-\frac{4}{3}x^{-\frac{4}{3}} - 3 + 8x$ **f** $5x^4 - x^{-\frac{3}{2}} - 12x^{-5}$

3 Find $f(x)$ when $f'(x)$ is given by the following expressions. In each case simplify your answer.

a $12x + \frac{3}{2}x^{-\frac{3}{2}} + 5$ **b** $6x^5 + 6x^{-7} - \frac{1}{6}x^{-\frac{7}{6}}$ **c** $\frac{1}{2}x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}}$

d $10x^4 + 8x^{-3}$ **e** $2x^{-\frac{1}{3}} + 4x^{-\frac{5}{3}}$ **f** $9x^2 + 4x^{-3} + \frac{1}{4}x^{-\frac{1}{2}}$

E/P **4** Find y given that $\frac{dy}{dx} = (2x + 3)^2$. **(4 marks)**

Problem-solving

Start by expanding the brackets.

E 5 Find $f(x)$ given that $f'(x) = 3x^{-2} + 6x^{\frac{1}{2}} + x - 4$. **(4 marks)**

Challenge

Find y when $\frac{dy}{dx} = (2\sqrt{x} - x^2)\left(\frac{3+x}{x^5}\right)$

13.2 Indefinite integrals

You can use the symbol $\int$ to represent the process of **integration**.

- $\int f'(x)dx = f(x) + c$

Notation This process is called **indefinite integration**. You will learn about **definite** integration later in this chapter.

You can write the process of integrating x^n as follows:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

The elongated S means integrate.

The expression to be integrated.

The dx tells you to integrate with respect to x.

When you are integrating a polynomial function, you can integrate the terms one at a time.

- $\int(f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$

Example 4

Find:

a $\int(x^{\frac{1}{2}} + 2x^3)\,dx$ **b** $\int(x^{-\frac{3}{2}} + 2)\,dx$ **c** $\int(p^2x^{-2} + q)\,dx$ **d** $\int(4t^2 + 6)\,dt$

a $\int(x^{\frac{1}{2}} + 2x^3)dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{2x^4}{4} + c$

First apply the rule term by term.

$= \frac{2}{3}x^{\frac{3}{2}} + \frac{1}{2}x^4 + c$

Simplify each term.

b $\int(x^{-\frac{3}{2}} + 2)dx = \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + 2x + c$

Remember $-\frac{3}{2} + 1 = -\frac{1}{2}$ and the integral of the constant 2 is $2x$.

$= -2x^{-\frac{1}{2}} + 2x + c$

c $\int(p^2x^{-2} + q)dx = \frac{p^2}{-1}x^{-1} + qx + c$

The dx tells you to integrate with respect to the variable x, so any other letters must be treated as constants.

$= -p^2x^{-1} + qx + c$

d $\int(4t^2 + 6)dt = \frac{4t^3}{3} + 6t + c$

The dt tells you that this time you must integrate with respect to t.

Use the rule for integrating x^n but replace x with t:

If $\frac{dy}{dt} = kt^n$, then $y = \frac{k}{n+1}t^{n+1} + c$, $n \neq -1$.

Before you integrate, you need to ensure that each term of the expression is in the form kx^n, where k and n are real numbers.

Example 5

Find:

a $\int\left(\frac{2}{x^3} - 3\sqrt{x}\right)\mathrm{d}x$ **b** $\int x\left(x^2 + \frac{2}{x}\right)\mathrm{d}x$ **c** $\int\left((2x)^2 + \frac{\sqrt{x}+5}{x^2}\right)\mathrm{d}x$

a $\int\left(\frac{2}{x^3} - 3\sqrt{x}\right)dx$

$= \int(2x^{-3} - 3x^{\frac{1}{2}})dx$ — First write each term in the form x^n.

$= \frac{2}{-2}x^{-2} - \frac{3}{\frac{3}{2}}x^{\frac{3}{2}} + c$ — Apply the rule term by term.

$= -x^{-2} - 2x^{\frac{3}{2}} + c$ — Simplify each term.

$= -\frac{1}{x^2} - 2\sqrt{x^3} + c$ — Sometimes it is helpful to write the answer in the same form as the question.

b $\int x\left(x^2 + \frac{2}{x}\right)dx$

$= \int(x^3 + 2)dx$ — First multiply out the bracket.

$= \frac{x^4}{4} + 2x + c$ — Then apply the rule to each term.

c $\int\left((2x)^2 + \frac{\sqrt{x}+5}{x^2}\right)dx$

$= \int\left(4x^2 + \frac{x^{\frac{1}{2}}}{x^2} + \frac{5}{x^2}\right)dx$ — Simplify $(2x)^2$ and write $\sqrt{x}$ as $x^{\frac{1}{2}}$.

$= \int(4x^2 + x^{-\frac{3}{2}} + 5x^{-2})dx$ — Write each term in the form x^n.

$= \frac{4}{3}x^3 + \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + \frac{5x^{-1}}{-1} + c$ — Apply the rule term by term.

$= \frac{4}{3}x^3 - 2x^{-\frac{1}{2}} - 5x^{-1} + c$ — Finally simplify the answer.

$= \frac{4}{3}x^3 - \frac{2}{\sqrt{x}} - \frac{5}{x} + c$

Exercise 13B

1 Find the following integrals:

a $\int x^3\,\mathrm{d}x$ **b** $\int x^7\,\mathrm{d}x$ **c** $\int 3x^{-4}\,\mathrm{d}x$ **d** $\int 5x^2\,\mathrm{d}x$

2 Find the following integrals:

a $\int(x^4 + 2x^3)\mathrm{d}x$ **b** $\int(2x^3 - x^2 + 5x)\mathrm{d}x$ **c** $\int(5x^{\frac{3}{2}} - 3x^2)\mathrm{d}x$

3 Find the following integrals:

a $\int(4x^{-2} + 3x^{-\frac{1}{2}})\mathrm{d}x$ **b** $\int(6x^{-2} - x^{\frac{1}{2}})\mathrm{d}x$ **c** $\int(2x^{-\frac{3}{2}} + x^2 - x^{-\frac{1}{2}})\mathrm{d}x$

4 Find the following integrals:

a $\int(4x^3 - 3x^{-4} + r)\mathrm{d}x$ **b** $\int(x + x^{-\frac{1}{2}} + x^{-\frac{3}{2}})\mathrm{d}x$

c $\int(px^4 + 2t + 3x^{-2})\mathrm{d}x$

Hint In **Q4** part **c** you are integrating with respect to x, so treat p and t as constants.

5 Find the following integrals:

a $\int(3t^2 - t^{-2})\mathrm{d}t$ **b** $\int(2t^2 - 3t^{-\frac{3}{2}} + 1)\mathrm{d}t$ **c** $\int(pt^3 + q^2 + px^3)\mathrm{d}t$

6 Find the following integrals:

a $\int\frac{(2x^3 + 3)}{x^2}\mathrm{d}x$ **b** $\int(2x + 3)^2\,\mathrm{d}x$ **c** $\int(2x + 3)\sqrt{x}\,\mathrm{d}x$

7 Find $\int f(x)\mathrm{d}x$ when $f(x)$ is given by the following:

a $\left(x + \frac{1}{x}\right)^2$ **b** $(\sqrt{x} + 2)^2$ **c** $\left(\frac{1}{\sqrt{x}} + 2\sqrt{x}\right)$

8 Find the following integrals:

a $\int\left(x^{\frac{2}{3}} + \frac{4}{x^3}\right)\mathrm{d}x$ **b** $\int\left(\frac{2 + x}{x^3} + 3\right)\mathrm{d}x$ **c** $\int(x^2 + 3)(x - 1)\mathrm{d}x$

d $\int\frac{(2x + 1)^2}{\sqrt{x}}\mathrm{d}x$ **e** $\int\left(3 + \frac{\sqrt{x} + 6x^3}{x}\right)\mathrm{d}x$ **f** $\int\sqrt{x}(\sqrt{x} + 3)^2\,\mathrm{d}x$

9 Find the following integrals:

a $\int\left(\frac{A}{x^2} - 3\right)\mathrm{d}x$ **b** $\int\left(\sqrt{Px} + \frac{2}{x^3}\right)\mathrm{d}x$ **c** $\int\left(\frac{p}{x^2} + q\sqrt{x} + r\right)\mathrm{d}x$

E **10** Given that $f(x) = \frac{6}{x^2} + 4\sqrt{x} - 3x + 2$, $x > 0$, find $\int f(x)\mathrm{d}x$. **(5 marks)**

E **11** Find $\int\left(8x^3 + 6x - \frac{3}{\sqrt{x}}\right)\mathrm{d}x$, giving each term in its simplest form. **(4 marks)**

E/P **12** **a** Show that $(2 + 5\sqrt{x})^2$ can be written as $4 + k\sqrt{x} + 25x$, where k is a constant to be found. **(2 marks)**

b Hence find $\int(2 + 5\sqrt{x})^2\,\mathrm{d}x$. **(3 marks)**

E **13** Given that $y = 3x^5 - \frac{4}{\sqrt{x}}$, $x > 0$, find $\int y\,\mathrm{d}x$ in its simplest form. **(3 marks)**

E/P **14** $\int\left(\frac{p}{2x^2} + pq\right)\mathrm{d}x = \frac{2}{x} + 10x + c$ **(5 marks)**

Find the value of p and the value of q.

Problem-solving

Integrate the expression on the left-hand side, treating p and q as constants, then compare the result with the right-hand side.

E/P **15** $f(x) = (2 - x)^{10}$

Given that x is small, and so terms in x^3 and higher powers of x can be ignored:

a find an approximation for $f(x)$ in the form $A + Bx + Cx^2$ **(3 marks)**

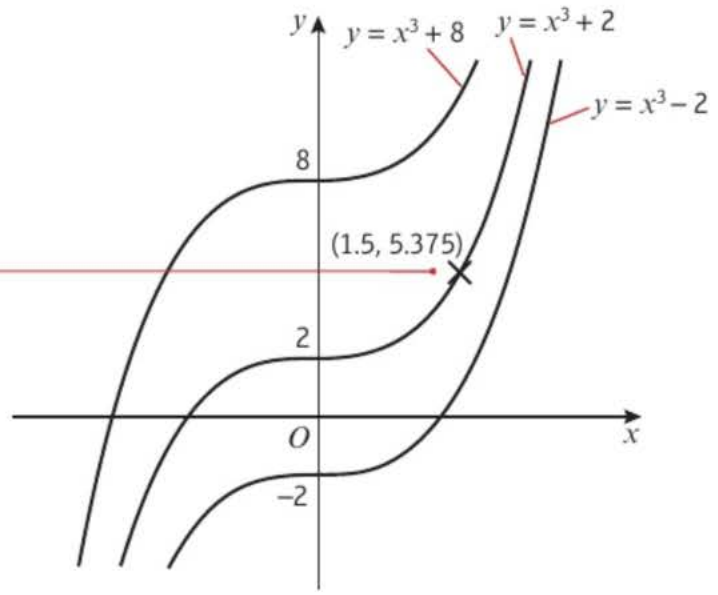
b find an approximation for $\int f(x)dx$. **(3 marks)**

Hint Find the first three terms of the binomial expansion of $(2 - x)^{10}$. ← Section 8.3

13.3 Finding functions

You can find the constant of integration, c, when you are given (i) any point (x, y) that the curve of the function passes through or (ii) any value that the function takes. For example, if $\frac{dy}{dx} = 3x^2$ then $y = x^3 + c$. There are infinitely many curves with this equation, depending on the value of c.

Only one of these curves passes through this point. Choosing a point on the curve determines the value of c.



- **To find the constant of integration, c**
  - **Integrate the function**
  - **Substitute the values (x, y) of a point on the curve, or the value of the function at a given point $f(x) = k$, into the integrated function**
  - **Solve the equation to find c**

Example 6

The curve C with equation $y = f(x)$ passes through the point (4, 5). Given that $f'(x) = \frac{x^2 - 2}{\sqrt{x}}$, find the equation of C.

$f'(x) = \frac{x^2 - 2}{\sqrt{x}} = x^{\frac{3}{2}} - 2x^{-\frac{1}{2}}$ — First write $f'(x)$ in a form suitable for integration.

So $f(x) = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} - \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + c$ — Integrate as normal and don't forget the + c.

$= \frac{2}{5}x^{\frac{5}{2}} - 4x^{\frac{1}{2}} + c$

But $f(4) = 5$ — Use the fact that the curve passes through (4, 5).

So $5 = \frac{2}{5} \times 2^5 - 4 \times 2 + c$

$5 = \frac{64}{5} - 8 + c$

$5 = \frac{24}{5} + c$

So $c = \frac{1}{5}$

So $y = \frac{2}{5}x^{\frac{5}{2}} - 4x^{\frac{1}{2}} + \frac{1}{5}$

Remember $4^{\frac{5}{2}} = 2^5$.

Solve for c.

Finally write down the equation of the curve.

Online Explore the solution using GeoGebra.



Exercise 13C

1 Find the equation of the curve with the given derivative of y with respect to x that passes through the given point:

a $\frac{dy}{dx} = 3x^2 + 2x$; point (2, 10)

b $\frac{dy}{dx} = 4x^3 + \frac{2}{x^3} + 3$; point (1, 4)

c $\frac{dy}{dx} = \sqrt{x} + \frac{1}{4}x^2$; point (4, 11)

d $\frac{dy}{dx} = \frac{3}{\sqrt{x}} - x$; point (4, 0)

e $\frac{dy}{dx} = (x + 2)^2$; point (1, 7)

f $\frac{dy}{dx} = \frac{x^2 + 3}{\sqrt{x}}$; point (0, 1)

2 The curve C, with equation $y = f(x)$, passes through the point (1, 2) and $f'(x) = 2x^3 - \frac{1}{x^2}$
Find the equation of C in the form $y = f(x)$.

3 The gradient of a particular curve is given by $\frac{dy}{dx} = \frac{\sqrt{x} + 3}{x^2}$. Given that the curve passes through the point (9, 0), find an equation of the curve.

E **4** The curve with equation $y = f(x)$ passes through the point $(-1, 0)$. Given that $f'(x) = 9x^2 + 4x - 3$, find $f(x)$. **(5 marks)**

E/P **5** $\frac{dy}{dx} = 3x^{-\frac{1}{2}} - 2x\sqrt{x}$, $x > 0$.
Given that $y = 10$ at $x = 4$, find y in terms of x, giving each term in its simplest form. **(7 marks)**

E/P **6** Given that $\frac{6x + 5x^{\frac{3}{2}}}{\sqrt{x}}$ can be written in the form $6x^p + 5x^q$,

a write down the value of p and the value of q. **(2 marks)**

Given that $\frac{dy}{dx} = \frac{6x + 5x^{\frac{3}{2}}}{\sqrt{x}}$ and that $y = 100$ when $x = 9$,

b find y in terms of x, simplifying the coefficient of each term. **(5 marks)**

(P) **7** The displacement of a particle at time t is given by the function $f(t)$, where $f(0) = 0$.
Given that the velocity of the particle is given by $f'(t) = 10 - 5t$,

a find $f(t)$

b determine the displacement of the particle when $t = 3$.

Problem-solving

You don't need any specific knowledge of mechanics to answer this question. You are told that the displacement of the particle at time t is given by $f(t)$.

(P) **8** The height, in metres, of an arrow fired horizontally from the top of a castle is modelled by the function $f(t)$, where $f(0) = 35$. Given that $f'(t) = -9.8t$,

a find $f(t)$.

b determine the height of the arrow when $t = 1.5$.

c write down the height of the castle according to this model.

d estimate the time it will take the arrow to hit the ground.

e state one assumption used in your calculation.

Challenge

1 A set of curves, where each curve passes through the origin, has equations $y = f_1(x)$, $y = f_2(x)$, $y = f_3(x)$ … where $f_n'(x) = f_{n-1}(x)$ and $f_1(x) = x^2$.

a Find $f_2(x)$, $f_3(x)$.

b Suggest an expression for $f_n(x)$.

2 A set of curves, with equations $y = f_1(x)$, $y = f_2(x)$, $y = f_3(x)$, … all pass through the point (0, 1) and they are related by the property $f_n'(x) = f_{n-1}(x)$ and $f_1(x) = 1$. Find $f_2(x)$, $f_3(x)$, $f_4(x)$.

13.4 Definite integrals

You can calculate an integral between two **limits**. This is called a **definite integral**. A definite integral usually produces a **value** whereas an indefinite integral always produces a **function**.

Here are the steps for integrating the function $3x^2$ between the limits $x = 1$ and $x = 2$.

The limits of the integral are from $x = 1$ to $x = 2$.

$$\left[\int_1^2 3x^2\,dx = [x^3]_1^2\right]$$

Write the integral in [ ] brackets.

$$= (2^3) - (1^3)$$

Write this step in ( ) brackets.

Evaluate the integral at the upper limit.

$$= 8 - 1$$

$$= 7$$

Evaluate the integral at the lower limit.